Ephemeral Memory

Successful Retention of Short-Term Information Using Variable Plasticity and Decaying Weight Values

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Memorizing a Phone Number

1. hear it, fresh memory in its entirety
2. keep the important details
3. write it down
4. repeat it (aloud or in your head)
5. wait, forget parts of it
6. review

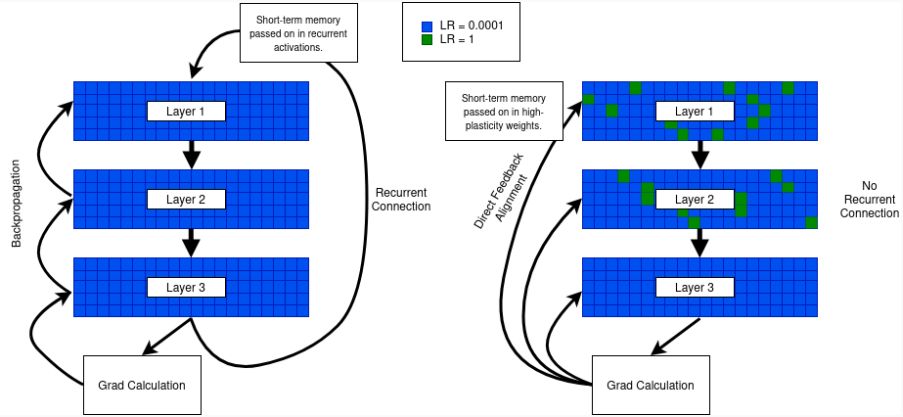
Existing Forms of “Memory”

human	NN Mechanism
hear it, fresh memory in its entirety	input activations, context window
keep the important details	attention mechanism, LSTM gate, small lr in pretraining
write it down	RAG, tool use
repeat it (aloud or in your head)	recurrent connection activation
wait, forget parts of it	weight decay, l2 norm
review	fine tuning, RL memory consolidation

Where It's Stored

human	NN Mechanism	Where It's Stored
hear it, fresh memory in its entirety	input activations, context window	input activations
keep the important details	attention mechanism, LSTM gate, small lr in pretraining	attention matrix (activations), recurrent activations
write it down	RAG, tool use	database
repeat it (aloud or in your head)	recurrent connection activation	activations
wait, forget parts of it	weight decay, l2 norm	weights
review	fine tuning, RL memory consolidation	weights!

Can Short-term Information be Stored in the Weights?



Experiments

Tasks:

- Repeated Sequences
- Key-Recall
- Reversed Sequences

Key-Recall examples:

```
0000?10!1
00000?200!2
00?,00!,
00?.0!.
```

Reversed examples:

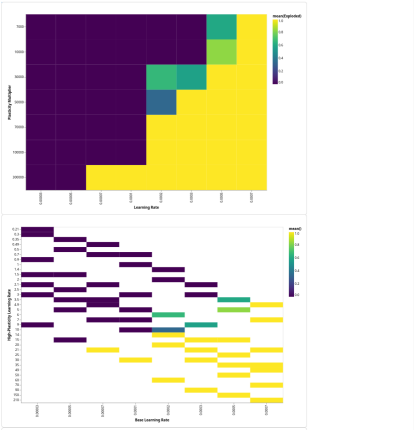
```
abcba
xyzyx
12321
qwerttrewq
```

Inherent Volatility

- Both **exploding gradients** and **ethereal-weight volatility** arise from **positive feedback** in the update rule.
- As **parameter magnitudes increase**, the **induced gradients** also grow, amplifying subsequent updates.
- Without **explicit damping** (e.g., weight decay, normalization, clipping), the **effective gain** exceeds 1, leading to runaway growth.

Stabilizing

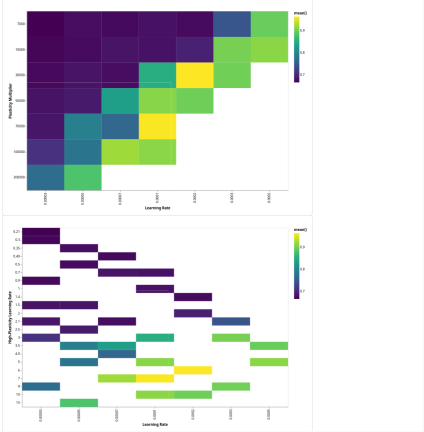
With weight decay and careful hyperparameter tuning, this can be stable!



Accuracy

While aggressive forgetting rates prevent explosion, they also reduce memory duration below useful thresholds.

Conversely, insufficient forgetting leads to inevitable instability. This creates a narrow operational window that varies significantly across tasks and datasets.



Multi-TImescale Learning

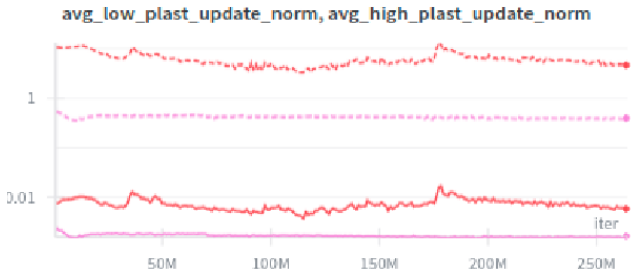


Figure 7: Gradient norm evolution throughout training shows clear separation between high-plasticity and low-plasticity weight categories. The high-plasticity weights (red) exhibit characteristic volatility with occasional spikes corresponding to memory encoding events, while low-plasticity weights (blue) maintain steady, controlled gradient magnitudes. This separation demonstrates that the two learning processes operate independently, allowing stable long-term learning to proceed alongside volatile working memory encoding.

The Control Problem

We want to control the plasticity in those fast weights:

$$x := \|w_{\text{eph}}\|.$$

Keep it in the corridor, so we can scale, fiddle with lr, whatever.

This is just a single scalar! Very basic info. (not a hessian)

Linearizing

- $\delta x \;=\; x - \bar{x}$ — state deviation
- $\delta u \;=\; \alpha - \alpha_0$ — control deviation (what the controller picks)
- $\delta d \;=\; g_{\text{raw}} - \bar{g}$ — disturbance deviation (what we can't control)

$$\delta x(t+1) \;=\; A\,\delta x(t) \;+\; B\,\delta u(t) \;+\; E\,\delta d(t).$$

A is $\partial(\text{update})/\partial x$ at the operating point: how a kick to the weight norm propagates one step forward. B is $\partial(\text{update})/\partial\alpha$: sensitivity of the next-step weight norm to the plasticity knob. E is $\partial(\text{update})/\partial g_{\text{raw}}$: sensitivity of the next-step weight norm to a fluctuation in gradient norm.

This all causes problems later bc A is time-varying.

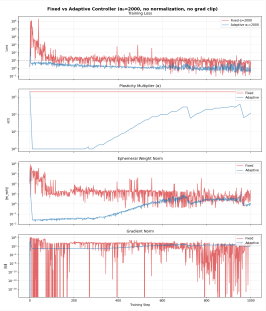
Attempts to Stabilize

Treat α as a control input and $\|w_{\text{eph}}\|$ as the state. Three controllers, all reading the live weight norm and choosing α at each character step:

Controller	Idea	Channel used
LQR	Solve discrete Riccati once, apply $u = -K\,\delta x$	A, B — ignores E by construction
H_∞	Minimize worst-case disturbance gain	A, B, E — full partitioned plant
Adaptive	Heuristic gain scheduling, updates K online from observed error	doesn't trust the linearization

LQR is the textbook starting point. H_∞ adds robustness against the gradient-norm disturbance. Adaptive is the one that doesn't take the LTI model too seriously and just reacts to what it sees.

Results — Palindrome Reversal



What Didn't Work

1. A **is time-varying.** The loss landscape morphs as training progresses; we identify A once and freeze it.
2. $\|w_{\mathbf{eph}}\|$ **isn't the state we actually care about.** Accuracy is not monotone in weight norm — you can keep the corridor and still lose the memory.
3. H_∞ **is conservative.** The worst-case budget gets spent on disturbances we couldn't actually afford to hedge against.

Where Next

- μ-**synthesis** to handle time-varying A as structured uncertainty
- **MPC** to express the corridor as a *hard constraint*, not a soft cost
- **Online identification** to track A as the loss landscape morphs
- **Vector-valued state** to capture the directional information lost when collapsing to $\|w_{\text{eph}}\|$
- **Connection to neuroscience**: short-term synaptic plasticity (STSP) has the same runaway-vs-decay tradeoff and may motivate the right state representation

Conclusion

- Ephemeral weights store short-term info, but the corridor between *fail-to-memorize* and *runaway* is narrow
- A linearized scalar plant captures the runaway dynamics with a single load-bearing parameter (the effective Hessian H)
- Adaptive gain scheduling beats fixed α; LQR and H_∞ work in narrower regimes
- Right next step: μ-synthesis with online identification